

Jujian Zhang

PERSONAL DETAILS

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EDUCATION

PhD. Mathematics 2021-Present
Imperial College, London
MA. Philosophy with Distinction 2020-2021
University College, London
MSc. Pure Mathematics with Distinction 2019-2020
Imperial College, London
BSc. Mathematics First Class Honours 2016-2019
University of Bristol

RELATED EXPERIENCE

Harmonic 2025 May - Present
Artificial Intelligence
NDA

Project Numina 2024-2025 April
Artificial Intelligence
Auto-formalisation of combinatorics problem at the difficulty of the International Mathematics Olympiad and other mathematical competitions.

Mathlib4 2020-Present
Open Source Project
I am an active contributor to Mathlib4, the largest mathematics formalisation project, which is utilized by artificial intelligence for proof searching.

Funded Study 2018
University of Bristol
In 2018 summer break, I studied some topics in topology and differential manifolds under supervision of Dr. Roman Schubert.

AWARD

Schrödinger Prize 2021
Imperial College London
Fully funded PhD studentship with London weighted stipend (about £60,000)

Faculty Prize Winner 2018
University of Bristol
Awarded to the highest performing undergraduate students at the end of the second year of BSc Mathematics

PUBLICATION

- Graded Rings in Lean's Dependent Type Theory** 2021
International Conference on Intelligent Computer Mathematics 2022
Lecture Notes in Computer Science
- Formalising the Proj Construction in Lean.** 2022
14th International Conference on Interactive Theorem Proving (ITP 2023)
Schloss Dagstuhl-Leibniz-Zentrum für Informatik, 2023.
- Formalisation of the Category of Hopf Algebras in Lean4** 2024
Mathematical Software – ICMS 2024 8th International Conference

OTHER WORKS AND WRITINGS

- A Formalisation of Transcendence of e** 2020
Some elementary result in transcendental number theory
- Identity Type under Homotopy Type Theory with Univalent Axiom** 2021
An alternative interpretation of identity type in homotopy type theory
- Formalising Flat Modules** 2022
Flat modules are of great importance in commutative algebra and algebraic geometry, this work formalised that many equivalent definitions of flatness are indeed equivalent.
- Dimension Theory** 2023-2024
Formalising Krull dimensions of rings and topological spaces by generalising this concept to an arbitrary binary relation and other related concepts such as lengths and depth of a module. This project is also related to flatness project by famous results such as the miracle flatness theorem.
- Brauer Group and Galois Cohomology** 2024
The group isomorphism between $\text{Br}(K/F) \cong H^2(\text{Gal}(K/F), K^)$. This project includes many results in classical noncommutative algebra, Morita theory and faithfully flat modules.*
- Multi-graded Proj Construction** 2025
A generalization of Proj construction for $\mathbb{N}$ -graded algebra to Proj construction for any M -graded algebra where M is an abelian group.